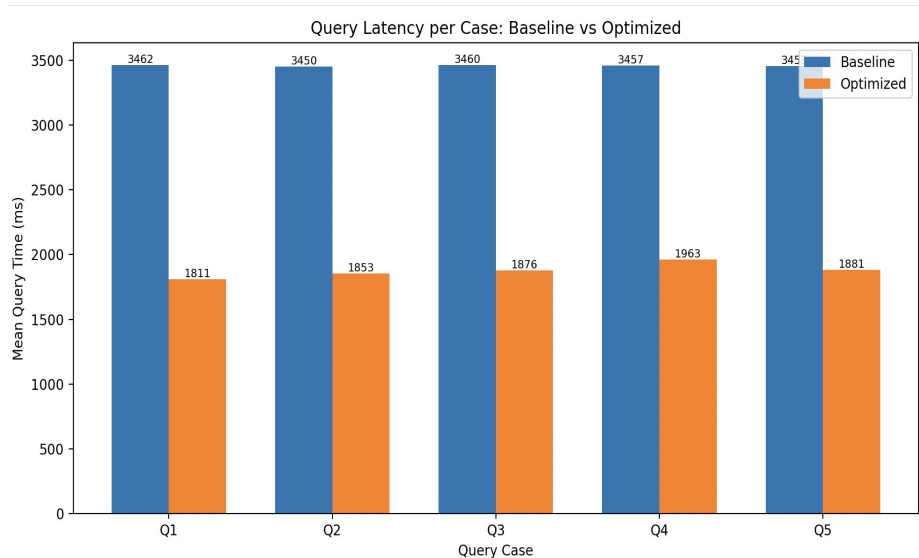


Distributed Query Processing: Fairness, Chunking, and Performance

• Meghana Koti, Brandon Llanes

CMPE 275



Client-observed Mean query fetch time — baseline (single large shard, precomputed) vs optimized (natural monthly shards, on-demand)

01 Shard Assignment Impacts Load Distribution

Weighted assignment by file size and trip count distributes work more evenly than round-robin.

02 Baseline vs Optimized: Server-Side Work

Baseline pre computed during SubmitQuery; optimized processed data on demand in bounded chunks.

03 Optimization Improved Client-Observed Time

On-demand processing with natural sharding cut mean fetch time by ~46% across all query cases.